

BenHalluEval: A Multi-Task Hallucination Evaluation Framework for Large Language Models on Bengali

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Abstract

Despite Bengali being the sixth most spoken language in the world, no prior work has systematically evaluated hallucination in large language models (LLMs) for Bengali. We introduce **BenHalluEval**, a hallucination evaluation framework covering four tasks: Question Answering (QA), Bangla-English Code-Mixed QA, Summarization, and Mathematical Reasoning. We construct 12,000 hallucinated candidates using GPT-5.4 across twelve hallucination types and evaluate nine LLMs under a dual-track protocol that independently measures false-positive rate on ground-truth instances (Track A) and hallucination detection rate on hallucinated candidates (Track B). Three native Bengali annotators validate the benchmark at almost-perfect agreement (Fleiss' $\kappa = 0.911\text{--}0.926$). To jointly penalise both failure modes, we propose **BenHalluScore**, a dual-track metric equivalent to balanced error rate. BenHalluScore ranges from 7.72% to 55.42%, with rankings task-conditional rather than scale-ordered. Chain-of-thought prompting lowers BenHalluScore in 14 of 32 evaluable combinations and raises it in 18, with gains on Summarization and Mathematical Reasoning but degradation on both QA variants. BenHalluEval establishes the first dedicated hallucination benchmark for Bengali and highlights the inadequacy of single-track and prompting-only approaches for low-resource settings. The dataset and code are available at <https://anonymous.4open.science/r/BanglaHalluEval-EB77>.

1 Introduction

Large language models (LLMs) generate outputs that contradict established facts, fabri-

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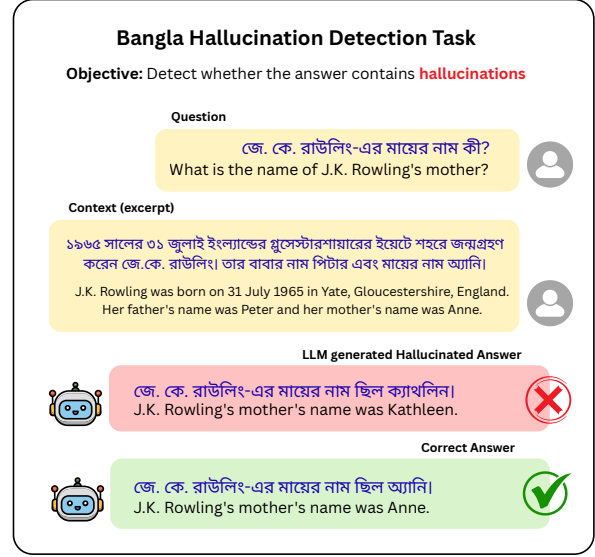


Figure 1: An example of hallucination detection in BenHalluEval.

cate details, or conflict with the provided context, known as hallucination (Huang et al., 2024; Li et al., 2023). A medical chatbot that invents drug dosages, a legal assistant that cites non-existent cases, or a tutoring system that presents wrong arithmetic can cause real harm. A model that scores well on standard Natural Language Understanding benchmarks can still hallucinate heavily in generation (Min et al., 2023). Without a comprehensive hallucination benchmark, there is no reliable way to assess how much an LLM can actually be trusted. As illustrated in Figure 1, given a Bengali question about J.K. Rowling's mother and a supporting context stating the answer is *Anne*, an LLM may instead generate *Kathleen*—a fabricated fact that contradicts the provided context which BenHalluEval is designed to detect.

Hallucination evaluation has been explored in different languages such as English (Li et al.,

2023) Arabic (Alansari and Luqman, 2025), Hindi and Turkish (Abdaljalil et al., 2026). Bengali, being the sixth most spoken language in the world (Sani et al., 2026), remains under-explored. Dedicated pretrained models (Bhattacharjee et al., 2022), multilingual benchmarks (Hasan et al., 2021), and task-specific datasets (Clark et al., 2020; Khan et al., 2023; Paul et al., 2025) already exist for Bengali, yet no benchmark measures hallucination in LLMs on both native Bengali and Bangla-English code-mixed input.

To fill this gap, we introduce **BenHalluEval**, a hallucination evaluation framework for Bengali covering four tasks: Question Answering (QA), Bangla-English Code-Mixed QA, Summarization, and Mathematical Reasoning. We use GPT-5.4¹ to generate hallucinated candidates, evaluate nine LLMs across three model categories, and systematically evaluate chain-of-thought prompting as a mitigation strategy across all model-task combinations. Our main contributions are:

- We present **BenHalluEval**, a hallucination evaluation framework for Bengali spanning four generation tasks for LLMs on Bengali and Bangla-English inputs, covering Question Answering (QA), Bangla-English code-mixed QA, summarization, and mathematical reasoning.
- We construct 12,000 hallucinated candidates using GPT-5.4 across task-specific hallucination categories, including a Bangla-English code-mixed setting, and evaluate them alongside ground-truth instances using an LLM-as-judge framework.
- We introduce **BenHalluScore** to jointly measure false positives and missed hallucinations, benchmark nine LLMs across three model categories, and systematically evaluate chain-of-thought prompting across all nine models and three tasks, showing the task-dependent patterns of improvement and degradation..

2 Related Work

Hallucination Evaluation Benchmarks. TruthfulQA, one of the earliest works on hallucination evaluation (Lin et al., 2022), showed

that model scale alone does not guarantee factual accuracy. FActScore advanced this by breaking long-form outputs into atomic claims for fine-grained precision measurement (Min et al., 2023). HaluEval established the sampling-then-evaluation paradigm across QA, dialogue, and summarization (Li et al., 2023). FaithBench (Bao et al., 2025) and HaluLens (Bang et al., 2025) subsequently refined faithfulness annotations and guarded against benchmark leakage. MedHallu extended this to the clinical domain, showing how domain specificity amplifies the impact of hallucination errors (Pandit et al., 2025). A shared limitation across these benchmarks is labelling bias where LLM judges tend to default to one verdict class regardless of content, a problem none of them directly measure (Li et al., 2023; Adib et al., 2025).

Low-Resource Hallucination and Bengali NLP. Hallucination rates scale inversely with pretraining data availability, with the gap widening in open-ended generation (Chataigner et al., 2024; Islam et al., 2025). AraHalluEval (Alansari and Luqman, 2025) evaluated Arabic-capable LLMs using fine-grained hallucination indicators, and Halluverse-M³ (Abdaljalil et al., 2026) extended multilingual evaluation to English, Arabic, Hindi, and Turkish. Neither benchmark incorporates dual-track calibration or examines code-mixed input. More recent work covers multi-domain understanding via BnMMLU (Joy and Shatabda, 2025), reasoning via MathMist (Sobhani et al., 2025), and summarization consistency via BanglaSummEval (Rafid et al., 2026). Bangla-English code-mixed usage is well-documented (Alam et al., 2025; Sheth et al., 2025; Alam et al., 2026), yet hallucination behaviour on Bangla-English input remains unexamined.

Positioning. We state our novelty claim precisely against three nearby lines of work. First, Bengali LLM evaluation that does not address hallucination, such as BenLLMEval (Kabir et al., 2024). Second, hallucination benchmarks built for other individual languages, such as AraHalluEval (Alansari and Luqman, 2025). Third, multilingual and fine-grained hallucination annotation benchmarks, such as ANAH-v2 (Gu et al., 2024), Hal-

¹<https://openai.com/index/introducing-gpt-5-4/>

luDial (Luo et al., 2024), and HalluVerse25 (Abdaljalil et al., 2025). Our claim is narrower than any of these: BenHalluEval is the first hallucination-*detection* benchmark built specifically for Bengali that covers both native-script Bengali and Bangla-English code-mixed input, and evaluates them under a dual-track protocol.

Mitigation Strategies. Chain-of-thought prompting (Wei et al., 2022) improves multi-step inference but can paradoxically obscure hallucination cues, making its benefit task- and model-dependent (Cheng et al., 2025). (Cheng et al., 2025) show that CoT reasoning can shift a model’s prediction patterns without improving its ability to distinguish hallucinated from correct content, with gains limited to tasks that require structured, verifiable intermediate steps.

BenHalluEval addresses this by covering four tasks: QA, Bangla-English Code-Mixed, QA, Summarization, and Mathematical Reasoning under a unified dual-track evaluation protocol.

3 Methodology

BenHalluEval is constructed through a three-stage pipeline: (i) seed dataset selection and sample filtering, (ii) hallucinated sample generation using GPT-5.4, and (iii) hallucination detection evaluation using nine LLMs. Figure 2 provides a full system overview. Each task has a task-specific selection strategy and hallucination categorization; the evaluation protocol is unified across all four tasks.

3.1 Stage 1: Seed Dataset Selection and Sample Filtering

Context-based Question Answering.

We use TyDiQA-GoldP (Bengali) (Clark et al., 2020), a dataset of 2,509 question–passage–answer triples. The supporting passage is provided to models at evaluation time, so a hallucination verdict reflects consistency with the passage rather than open-domain world knowledge. To select the samples that are more prone to hallucination, we prompted three models zero-shot with only the question, without the passage: Qwen2.5-

14B², Gemma-2-9B³, and DeepSeek-R1-14B⁴. A Qwen2.5-32B-Instruct⁵ judge scores each response as correct or incorrect. Samples that all three models answer incorrectly form the hard subset, yielding 1,313 instances (52.33%), from which 1,000 are randomly selected as the QA seed dataset. These three models were chosen because they represent three distinct model families, fit within our available computational resources (a single RTX 3090 with 24GB VRAM), and have been documented to perform competitively on Bengali tasks (Bhowmik et al., 2025; Chae et al., 2026). Since an instance is kept only if *all three* fail, no single model’s weakness determines the pool. To rule out same-family bias in the Qwen judge, we re-judged a random 500-instance subsample (seed 42) with GPT-4.1 mini: the judges agree on 95.2%, 92.4%, and 91.2% of Qwen2.5-14B, DeepSeek-R1-14B, and Gemma-2-9B answers respectively, with disagreement *lowest* for the same-family generator (Appendix J). Table 1 shows per-model zero-shot accuracy.

Model	Incorrect	Acc. (%)
DeepSeek-R1-14B	1,747	30.37
Gemma-2-9B	1,668	33.54
Qwen2.5-14B	1,948	22.36
All 3 incorrect	1,313	52.33

Table 1: Zero-shot performance of three models on TyDiQA-GoldP (Bengali, 2,509 instances).

Bangla-English Code-Mixed QA.

Bangla-English code-mixed text is widely used by Bengali speakers on social media and messaging platforms (Alam et al., 2025; Sheth et al., 2025; Alam et al., 2026). To evaluate hallucination detection in this setting, we take the same 1,000 QA samples and convert the questions from Bengali to Bangla-English code-mixed using GPT-5.4 (see Appendix D.5). Original answers are kept in Bengali. No extra filtering is needed, since these samples are already the hard subset from the QA selection. To verify conversion quality, we manually reviewed 100 randomly

²<https://huggingface.co/Qwen/Qwen2.5-14B>

³<https://huggingface.co/google/gemma-2-9b>

⁴<https://ollama.com/library/deepseek-r1:14b>

⁵<https://huggingface.co/Qwen/Qwen2.5-32B-Instruct>

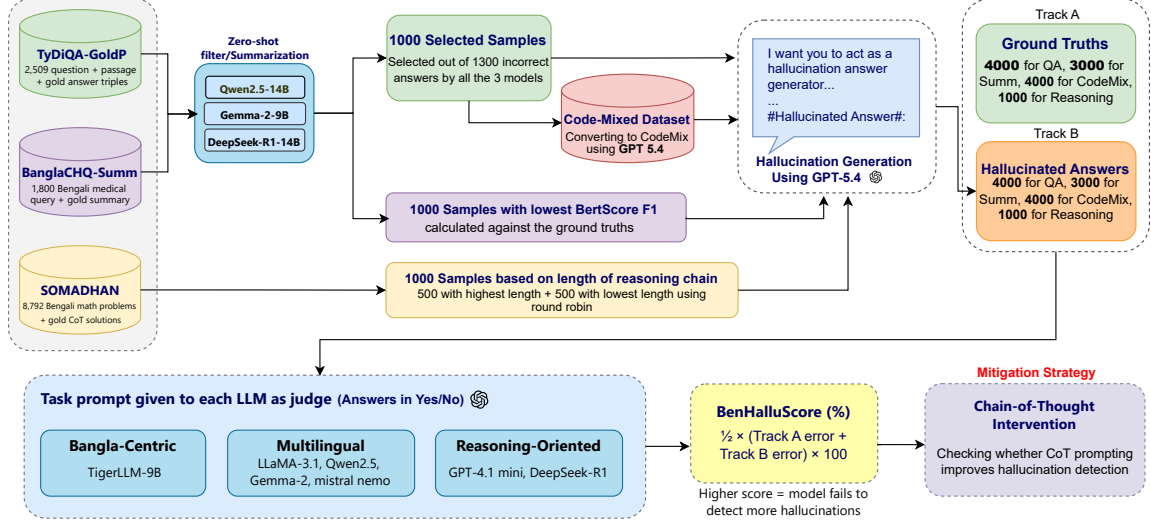


Figure 2: BenHalluEval system pipeline. Four tasks (QA, Code-Mixed QA, Summarization, Mathematical Reasoning) share a common evaluation stage. Track A evaluates ground-truth correct answers; Track B evaluates hallucinated candidates.

selected samples. All authors are native Bengali speakers, and the reviewed samples were found to be natural and linguistically appropriate.

Summarization. We use BanglaCHQ-Summ (Khan et al., 2023), a dataset of 1,800 Bengali consumer health query–summary pairs. We filter the samples based on semantic similarity using BERTScore F1. The same three models generate zero-shot summaries for all 1,800 samples, which are then scored against the original summaries using BERTScore F1 (Zhang et al., 2020). The average BERTScore F1 is computed per sample across all three models. The 1,000 samples with the lowest average scores are selected, as these are the samples that all three models struggle most to summarize faithfully. We use this only as a Stage 1 difficulty filter, not a faithfulness signal. Appendix J.2 confirms that our seed selection criterion (BERTScore similarity) is independent of faithfulness (Spearman $\rho = -0.06$), while an NLI metric cleanly separates our hallucinated candidates from gold summaries (AUC = 0.90), confirming the generated violations are genuine.

Mathematical Reasoning. We use SOMADHAN (Paul et al., 2025), a dataset of 8,792 Bengali math word problems with manually written chain-of-thought (CoT) solutions.

Samples are ranked by the length of their reasoning chain. The 500 longest and 500 shortest reasoning chains are selected as the two sampling pools. For each of the five error types, 100 samples with the longest reasoning chains and 100 with the shortest are selected, yielding 1,000 samples in total (200 per type). To ensure equal coverage, samples are picked alternately from the long and short pools in a fixed round-robin order, so that each error type receives an equal number of long and short reasoning chains. Chain length correlates with reasoning error patterns in large reasoning models (Zeng et al., 2025); Appendix J.3 confirms this.

3.2 Stage 2: Hallucinated Sample Generation

All hallucinated samples are generated using GPT-5.4. Each prompt specifies the hallucination type, includes an example of hallucination, and formats the target sample with field markers.

QA Hallucination Generation. Four hallucination types are applied to each of the 1,000 QA seed samples, one API call per type (full prompt is in Appendix D.2). This generates 4,000 hallucinated candidates in total:

- **Factualness:** Fabricates a concrete fact such as a name, date, or place that contradicts the context.

- **Comprehension:** Misunderstands the question’s context or intention and answers a different question.
- **Specificity:** Answers at an inappropriate level of detail, either too vague or too specific.
- **Inference:** Draws a conclusion that cannot be logically inferred from the given information.

Bangla-English Code-Mixed QA Hallucination Generation. The same four hallucination types used for QA are applied to the 1,000 Bangla-English Code-mixed seed samples. The generation process mirrors the QA pipeline exactly, with one difference: both the prompt and the generated hallucinated answers are in Bangla-English (Roman-script Bengali) rather than native Bengali script (full prompt is in Appendix D.2).

Summarization Hallucination Generation. Three hallucination types are applied to each of the 1,000 summarization seed instances, generating 3,000 hallucinated summaries in total:

- **Fabricated Content:** Introduces a detail that does not exist anywhere in the source document.
- **Non-Factual Addition:** Inserts a plausible but unverifiable claim not supported by the source.
- **Direct Contradiction:** Reverses information that is explicitly stated in the source.

Full prompt is in Appendix D.3.

Reasoning Hallucination Generation. We apply five error types, one per sample, selecting 1,000 hallucinated CoT chains. GPT-5.4 is prompted to modify exactly one reasoning step per instance and return output as a structured JSON object (full prompt in Appendix D.4):

- **Arithmetic Slip:** Silently alters a single numerical computation by a small amount.
- **Formula Misapplication:** Substitutes a related but incorrect mathematical operation.
- **Invalid Deduction:** Injects a logically unsupported transition between two reasoning steps.

- **Hallucinated Intermediate Fact:** Introduces a fabricated assumption mid-chain as if it were given.
- **Semantic Drift:** Reinterprets the question mid-reasoning so the final answer addresses a different goal.

Hallucination types are assigned by construction - one type per generation call which are adapted from HaluEval’s taxonomy (Li et al., 2023). So labelling is single-label throughout. Category boundaries for conceptually adjacent types are detailed in Appendix B.

3.3 Stage 3: Hallucination Detection Evaluation

Evaluation tracks. Evaluating only on hallucinated candidates risks rewarding models that blindly predict “Yes” for every input, while evaluating only on ground-truth samples risks rewarding models that always predict “No”. To get a more accurate picture of model behaviour, we evaluate each task on two separate tracks. **Track A** presents 1,000 ground-truth correct answers, where the expected verdict is “No” (not hallucinated). **Track B** presents the hallucinated candidates, where the expected verdict is “Yes”. This separation allows hallucination detection rate and false-positive rate to be measured independently.

To measure performance across both tracks jointly, we introduce **BenHalluScore** (BHS). A good model must both detect hallucinated content and avoid flagging correct answers as hallucinated. Neither track alone captures this. BenHalluScore penalises both failure modes equally, so a higher score indicates worse overall calibration. Formally:

$$\text{BenHalluScore} = \frac{1}{2} \left(\frac{W_A}{N_A} + \frac{W_B}{N_B} \right) \times 100\% \quad (1)$$

Relation to existing metrics. BenHalluScore equals Balanced Error Rate (1 – Balanced Accuracy): a model that answers “Yes” to everything scores 50%, not the $F_1 = 0.89$ it would receive on our QA split. Probabilistic metrics (ECE, Brier score) do not apply since the protocol elicits a binary verdict, not a confidence estimate. Equal weighting is

I want you act as a hallucination answer generator. The answer should be given in BANGLA . Given a question, right answer, and related knowledge, your objective is to write a hallucinated answer that sounds plausible but is factually incorrect. You SHOULD write the hallucinated answer using the following method:
1. You are trying to answer a question but there is a factual contradiction between the answer and the knowledge. You can fabricate some information that does not exist in the provided knowledge. (Demonstrations)
2. You are trying to answer a question but you misunderstand the question context and intention. (Demonstrations)
3. You are trying to answer a question but the answer is too general or too specific to answer the question at an appropriate level of specificity. (Demonstrations)
4. You are trying to answer a question but the answer cannot be inferred from the knowledge. You can incorrectly reason with the knowledge to arrive at a hallucinated answer. (Demonstrations)
You should try your best to make the answer become hallucinated. #Hallucinated Answer# can only have about 5 more words than #Right Answer# . #Knowledge#: <i><insert the related knowledge/context></i> #Question#: <i><insert the question></i> #Right Answer#: <i><insert the right answer to the question></i> #Hallucinated Answer#: Generate

Table 2: Hallucination generation prompt for the QA task. Four methods correspond to factual contradiction, comprehension mismatch, inappropriate specificity, and incorrect inference errors. Each instance generates one hallucinated answer per method under controlled constraints. Full prompt with Bengali demonstrations is in Appendix 9.

Task	Types	Seed	Hallu.	Gold
QA	4	1,000	4,000	1,000
Code-Mixed QA	4	1,000	4,000	1,000
Summarization	3	1,000	3,000	1,000
Math. Reasoning	5	1,000	1,000	1,000
Total			12,000	4,000

Table 3: BenHalluEval benchmark statistics. “Hallu.” = hallucinated candidates generated by GPT-5.4; “Gold” = ground-truth correct instances (Track A evaluation).

the unique symmetric setting under which neither degenerate responder wins. Appendix G reports sensitivity to other weightings.

Human validation. Three native Bengali-speaking annotators independently labelled all the instances spanning all four tasks and twelve hallucination types (Appendix I), blind to type and source, reaching almost-perfect agreement (Fleiss’ $\kappa = 0.911$ – 0.926 ; exact agreement 97–98%). Code-mixed conversions were validated separately for meaning preservation ($\kappa = 0.906$) and naturalness (4.38/5).

where W_A and W_B are the number of wrong verdicts on Track A and Track B respectively, N_A and N_B are the total instances per track, and the factor $\frac{1}{2}$ normalises the combined error to the $[0, 100]$ range. A score of 0%

indicates perfect calibration on both tracks; 50% indicates random or uniformly biased behaviour; 100% indicates perfectly wrong predictions across both tracks.

4 Experimental Setup

Models. We evaluate nine LLMs spanning three categories (Table 5). The reasoning-oriented category comprises DeepSeek-R1-14B and GPT-4.1 mini⁶. The multilingual category includes Qwen2.5-32B-Instruct, Gemma-2-27B⁷, Mistral-nemo-12B⁸, and LLaMA-3.1-8B-Instruct⁹. The Bangla-centric category comprises TigerLLM-9B (Raihan and Zampieri, 2025), TituLLM-3B (Nahin et al., 2025), and BanglaLLaMA-13B (Zehady et al., 2026). We deliberately include three systems here rather than one so that category-level statements rest on more than a single model; all three are released with published Bengali benchmarking, which lets us attribute differences within the category to factors other than exposure to Bengali text. Note that DeepSeek-R1-14B appears both in Stage 1 seed filtering

⁶<http://developers.openai.com/api/docs/models/gpt-4.1-mini>

⁷<https://huggingface.co/google/gemma-2-27b>

⁸<https://mistral.ai/news/mistral-nemo>

⁹<https://huggingface.co/meta-llama/Llama-3.1-8B-Instruct>

Model	QA			Code-Mixed QA			Summarization			Math. Reasoning		
	A-err.	B-err.	BHS	A-err.	B-err.	BHS	A-err.	B-err.	BHS	A-err.	B-err.	BHS
DeepSeek-R1-14B	48.00	29.20	38.60	40.10	13.78	26.94	17.60	6.73	12.17	19.90	16.30	18.10
GPT-4.1 mini	13.90	17.23	15.56	24.00	19.80	21.90	8.20	11.37	9.79	31.70	21.10	26.40
Qwen2.5-32B	27.90	65.80	46.85	17.40	30.60	24.00	4.70	10.73	7.72	24.80	28.80	26.80
Gemma-2-27B	16.40	78.50	47.45	19.80	38.15	28.98	4.20	31.40	17.80	5.50	72.60	39.05
Mistral-nemo-12B	87.70	19.98	53.84	85.90	24.93	55.42	70.50	9.87	40.19	1.60	97.00	49.30
LLaMA-3.1-8B	13.80	81.63	47.71	13.80	78.05	45.93	31.80	25.77	28.79	17.40	89.20	53.30
TigerLLM-9B	18.40	38.17	28.28	25.80	37.28	31.54	4.80	47.53	26.17	21.40	85.50	53.45
TituLLM-3B	99.75	0.25	50.00	99.25	2.00	50.62	100.00	0.33	50.16	100.00	0.00	50.00
BanglaLLaMA-13B	83.25	14.00	48.62	85.50	11.25	48.38	89.00	4.67	46.84	100.00	0.00	50.00

Table 4: BenHalluScore results across all four tasks and nine models (all values in %). A-err.: Track A error rate on ground-truth instances. B-err.: Track B error rate on hallucinated candidates. green cell = best (lowest) BHS per task; red cell = worst (highest) BHS per task. Full per-task results with raw counts in Appendix F.

and in the evaluated set. It was *not* used for hallucination generation, which was performed exclusively with GPT-5.4, so no evaluated model is scored on candidates it authored; Stage 1 only determines which seed instances enter the benchmark (Section 3).

Category	Model	Params
Reasoning	DeepSeek-R1-14B	14B
	GPT-4.1 mini	—
Multilingual	Qwen2.5-32B-Instruct	32B
	Gemma-2-27B	27B
	Mistral-nemo-12B	12B
	LLaMA-3.1-8B-Instruct	8B
Bangla-centric	TigerLLM-9B	9B
	TituLLM-3B	3B
	BanglaLLaMA-13B	13B

Table 5: Nine LLMs evaluated in BenHalluEval, grouped by model category.

Inference Settings. All models produce binary verdicts (*yes/no*) at temperature = 0, with a maximum output length of 1,024 tokens. Each instance is presented with its associated context - the passage for QA and Bangla-English code-mixed QA, the source article for summarisation, and the problem statement for reasoning. QA, summarisation, and reasoning prompts are issued in Bengali and the code-mixed prompts are in Bangla-English. GPT-4.1 mini is accessed via the OpenAI API. All remaining models are served locally using Ollama on a PC equipped with 128 GB

RAM and an NVIDIA RTX 3090 GPU (24 GB VRAM). Full evaluation prompts are provided in Appendix E.

5 Results Analysis

5.1 Question Answering

Table 4 shows QA results under the dual-track BenHalluScore. GPT-4.1 mini achieves the best BenHalluScore (15.56%), maintaining low errors on both tracks (A-err. 13.90%, B-err. 17.23%) - the only model that performs well simultaneously on both. DeepSeek-R1-14B scores 38.60%, with a notably high Track A error of 48.00%, meaning it incorrectly flags nearly half of all correct answers as hallucinated. Mistral-nemo-12B scores the worst (53.84%), driven almost entirely by its 87.70% Track A error - it flags nearly every ground-truth answer as hallucinated. LLaMA-3.1-8B shows the opposite pattern, predicting hallucination on 81.63% of Track B instances while maintaining a low Track A error (13.80%), reflecting opposing systematic biases rather than content-sensitive judgements. Notably, TigerLLM-9B, the only Bengali-centric model evaluated, achieves the second-best BenHalluScore on QA (28.28%), outperforming significantly larger multilingual models including Qwen2.5-32B, Gemma-2-27B, and even the reasoning-oriented DeepSeek-R1-14B. This is not, however, a property of Bengali pretraining as such: the two other Bangla-centric mod-

els in our suite collapse to near-uniform verdicts on the same task.

5.2 Bangla-English Code-Mixed QA

Bangla-English code-mixed results differ from QA for several models (Table 4), showing that script condition interacts with model behaviour. GPT-4.1 mini remains the best (BHS 21.90%), and Mistral-nemo-12B (55.42%) and LLaMA-3.1-8B (45.93%) retain their characteristic biases in the Roman-script condition. However, Qwen2.5-32B, Gemma-2-27B, and DeepSeek-R1-14B improve sharply on Bangla-English code-mixed relative to QA (by 22.85, 18.47, and 11.66 BHS points respectively), driven primarily by lower Track A errors. A plausible explanation is that Roman-script input shifts these models toward their stronger English-anchored representations, producing more consistent verdicts than native Bengali script - the opposite of the intuitive expectation that code-mixing would degrade performance.

5.3 Summarization

Across all models, BenHalluScores on summarization are notably lower than on QA (Table 4), suggesting that access to a full source article provides stronger contextual basis for factual consistency judgements. Qwen2.5-32B achieves the lowest (best performing) BenHalluScore (7.72%), with low errors on both tracks (A-err. 4.70%, B-err. 10.73%), while DeepSeek-R1-14B ranks second (12.17%) despite recording the lowest Track B error (6.73%). Mistral-nemo-12B again scores worst (40.19%), driven by a 70.50% Track A error. TigerLLM-9B shows the opposite pattern - a 47.53% Track B error against a 4.80% Track A error (BHS 26.17%) - indicating that labelling bias persists even when the model has access to sufficient context.

5.4 Mathematical Reasoning

Mathematical reasoning is the hardest of the four tasks in the sense that matters here: its median BenHalluScore is 49.30%, so half the evaluated models perform at or worse than a uniform responder that ignores content entirely (Table 4). DeepSeek-R1-14B achieves the best and most balanced result (A-err. 19.90%, B-err. 16.30%, BHS 18.10%).

At the other end, Mistral-nemo-12B predicts “No” for nearly every instance regardless of content (A-err. 1.60%, B-err. 97.00%, BHS 49.30%), a pattern indicating uniform label collapse. LLaMA-3.1-8B and TigerLLM-9B occupy the opposite extreme with Track B errors of 89.20% and 85.50% respectively, resulting in the worst BenHalluScores (53.30% and 53.45%). The concentration of models near or above 50% BenHalluScore indicates that reasoning hallucination in Bengali is particularly difficult to detect.

Chain-of-Thought Mitigation. We apply CoT to all nine models across all four tasks. BanglaLLaMA-13B produced no extractable verdict under CoT on any task, returning repetition-collapse output instead. Its four zero-shot scores are retained in Table 6 but excluded from the After-CoT analysis, leaving 32 evaluable After-CoT combinations across the remaining eight models. CoT lowers BenHalluScore in 14 of these 32 and raises it in 18. Since BenHalluScore is a balanced error rate, every reduction is by construction an improvement in discrimination, but magnitude varies sharply. Only four combinations improve by more than five points and finish at or below 30% fac TigerLLM-9B Reasoning (53.45→30.20), Mistral-nemo-12B Summarization (40.19→29.75), TigerLLM-9B Summarization (26.17→17.99), and GPT-4.1 mini Reasoning (26.40→18.45) — and exactly one improves on *both* tracks at once (TigerLLM-9B Reasoning). Task governs the pattern more than model family: Summarization improves for 5 of 8 models and Reasoning for 4 of 8, while QA degrades for 6 of 8 and Code-Mixed QA for 5 of 8, consistent with Cheng et al. (2025)’s finding that CoT gains require verifiable intermediate steps. DeepSeek-R1-14B, which has built-in CoT, degrades on all four tasks; GPT-4.1 mini, the strongest zero-shot QA model, suffers the largest single degradation in the grid (+21.22 on QA).

6 Discussion

Necessity of dual-track calibration. Detection-oriented benchmarks are conventionally scored on hallucinated candidates alone (Li et al., 2023; Gu et al., 2024; Luo et al., 2024; Abdaljalil et al., 2025), leaving un-

examined whether a model also recognises correct output as non-hallucinated. We note AraHalluEval (Alansari and Luqman, 2025) evaluates output factuality rather than detection ability, so it is not directly comparable. Our contrast is with single-track detection protocols. BenHalluEval’s Track A exposes failures Track B alone would conceal - Mistral-nemo-12B is competitive on Track B yet collapses on Track A - and the CoT results reinforce this: several reductions in Track B error coincide with Track A error rising above 90%, a trade BenHalluScore records as roughly neutral but a single-track metric would call progress.

QA vs. Code-Mixed QA. The Bangla-English Code-Mixed QA task shares its hallucination taxonomy and candidate instances with QA, differing only in question script (Bengali to Bangla-English). Despite this structural equivalence, BenHalluScores diverge across both tasks (Table 4). Most multilingual models achieve better results on Bangla-English Code-Mixed QA, whereas TigerLLM-9B and GPT-4.1 mini perform more favourably on Bengali QA. This indicates that the input script has a measurable influence on hallucination judgement even when the underlying content is identical.

Bengali-centric pretraining is not sufficient on its own. Expanding the Bangla-centric category from one model to three does not support our earlier claim that Bengali pretraining explains TigerLLM-9B’s calibration advantage: TituLLM-3B and BanglaLLaMA-13B, also Bangla-centric, sit at or near 50% BenHalluScore on every task via near-uniform verdicts (Table 4). We therefore retract the stronger claim: parameter count is a poor predictor of Bengali hallucination calibration, but Bengali-centric pretraining is neither necessary nor sufficient for it - TigerLLM-9B’s advantage more plausibly reflects instruction tuning and prompt compatibility.

What separates strong from weak detectors. Three factors dominate. First, **verdict-format compliance**: TituLLM-3B and BanglaLLaMA-13B score near 50% by emitting a near-constant label rather than misjudging content. Second, **evidence structure**: Summarization, with the longest ex-

plicit evidence span, yields the lowest BenHalluScore for 8 of 9 models; Mathematical Reasoning, requiring derivation rather than lookup, has the highest median (49.30%). Third, **script anchoring**: three multilingual models improve by up to 22.85 points on Code-Mixed QA relative to QA despite identical content, while TigerLLM-9B and GPT-4.1 mini move the other way. Parameter count predicts little.

Weighting robustness. We vary the relative weight of Track A and Track B errors across nine settings (Appendix G). Model rankings stay stable in the middle range. Only at the extremes does a degenerate responder win - TituLLM-3B, which detects almost no hallucinations, tops three of four tasks when Track A is weighted at 0.1. Equal weighting is the only setting where no such responder can benefit.

Limits of prompting-based mitigation. CoT lowers BenHalluScore in 14 of 32 evaluable combinations and raises it in 18, governed more by task than model family: Summarization and Reasoning account for nine of the fourteen gains, while QA degrades for 6 of 8 models and Code-Mixed QA for 5 of 8, consistent with Cheng et al. (2025). Half of the gains land at or above 44% BHS, so the model remains largely uninformative even after improving. Models with built-in CoT do not benefit from external CoT instruction: DeepSeek-R1-14B degrades on all four tasks, and the strongest zero-shot model, GPT-4.1 mini, suffers the largest single degradation in the grid.

7 Conclusion

Reliable hallucination evaluation for Bengali has remained absent despite the language’s global reach. We introduced BenHalluEval, a benchmark spanning four Bengali and Bangla-English generation tasks with 12,000 hallucinated candidates and 4,000 gold instances, evaluated across nine LLMs under a dual-track protocol and validated by three native-speaker annotators at almost-perfect agreement (Fleiss’ $\kappa = 0.911\text{--}0.926$). We proposed BenHalluScore, a balanced dual-track measure that penalises false alarms on correct output and missed hallucinations equally

and so cannot be gamed by uniform response bias. BenHalluScore ranges from 7.72% to 55.42% across models and tasks, and rankings are task-conditional rather than scale-ordered: parameter count predicts calibration poorly, and Bengali-centric pretraining is neither necessary nor sufficient for it. Chain-of-thought prompting lowers BenHalluScore in 14 of 32 evaluable combinations and raises it in 18, with gains clustering on summarization and mathematical reasoning while both question-answering variants degrade predominantly. Prompting-only mitigation is therefore insufficient for Bengali.

Limitations

While we conduct manual validation for selected components of the benchmark, the hallucinated candidates are generated automatically using GPT-5.4 rather than fully authored by human annotators. As a result, the benchmark may still reflect generation-specific patterns that differ from naturally occurring hallucinations in deployed systems. We are extending candidate construction to a second generator from a different family to quantify this. BenHalluScore assigns equal weight to errors on ground-truth and hallucinated tracks to reduce the influence of systematic response tendencies, but deployment settings with asymmetric costs may require task-specific weighting; Appendix G reports the full sensitivity analysis. All models are evaluated zero-shot at temperature 0, so results may differ under few-shot prompting or model-specific prompt tuning. Two of the nine models (TituLLM-3B, BanglaLLaMA-13B) were evaluated on a stratified sample rather than the full benchmark, and BanglaLLaMA-13B could not be evaluated under CoT at all. The chain-of-thought mitigation experiment now covers all four tasks across eight models; the interaction between code-mixed input and step-by-step prompting is examined in Table 6. Finally, BenHalluEval is scoped to four Bengali and Bangla-English task settings; generalization to other low-resource languages, Bengali genres, domains, and more complex mathematical reasoning remains to be validated.

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A Annotation Protocol and Guidelines

Annotators. Three undergraduate annotators, all native Bengali speakers with written and spoken Bangla coursework to at least the intermediate level. Annotators were recruited from within the authors’ institution, were informed of the research purpose, and consented to the use of their annotations.

Task presented. For each item the annotator saw the source context (passage, document, or problem statement), the question or prompt where applicable, and one candidate response, and answered a single binary question: *does the response contain information that is not supported by, or that contradicts, the given context?* Annotators did not see the hallucination type, whether the item came from the gold or the generated pool, or any model verdict. Item order was randomised and gold and hallucinated items were interleaved in benchmark proportions.

Decision criteria. (i) Judge only against the supplied context, never against outside world knowledge. (ii) A response that is incomplete but contains nothing unsupported is *not* hallucinated. (iii) For summaries, an inference the document plainly licenses is not hallucinated; an inference requiring an unstated premise is. (iv) For reasoning chains, a chain is hallucinated if any step is arithmetically wrong, uses an unstated quantity, or answers a different question than the one posed, even

when the final answer is numerically correct. (v) Fluency, register, and code-mixing style are never grounds for a hallucination verdict.

Adjudication. Ground truth was set by majority vote (2 of 3); no single annotator could determine a label. Items on which an annotator abstained were excluded rather than imputed; nine QA items were excluded on this basis.

Code-mixed conversion review. The same annotators rated each sampled conversion on two independent axes: meaning preservation relative to the Bengali original (binary) and naturalness as Bangla-English written by a native speaker (1–5 Likert). Items unanimously flagged as not preserving meaning were revised or excluded from the benchmark.

B Hallucination Type Definitions and Examples

Table 7 lists all hallucination types used across the three generation tasks, with a definition and a translated English example for each. Bangla-English Code-Mixed QA uses the same four types as QA, so we do not repeat them here.

Labelling protocol. Each candidate is produced under a prompt targeting exactly one hallucination type (Appendix D), so no multi-label adjudication arises. Conceptually adjacent categories remain separable by their generative constraint: *Fabricated Content* introduces a detail with no referent in the source, whereas *Non-Factual Addition* elaborates an entity that *is* present in the source with an unverifiable claim; *Arithmetic Slip* preserves the operator and perturbs only the operand, whereas *Formula Misapplication* preserves the operands and substitutes the operator.

Why *Inference* is a distinct QA category. Because the supporting passage is always provided, all four QA types are violations *relative to that passage*; they differ in how the violation arises. *Factualness* requires the passage to state a competing value for the same slot, which the hallucination replaces. *Inference* requires the passage to state *no* value for that slot, and the hallucination supplies one by an unlicensed step: a passage locat-

ing Cox’s Bazar in Chittagong Division neither states nor entails which city is the capital, so “Cox’s Bazar is the capital” is unsupported rather than contradicted. The operational test used during generation is whether the hallucinated span could be deleted without changing the passage’s truth conditions — it could not for *Factualness*, it could for *Inference*.

Semantic Drift vs Hallucinated Intermediate Fact. A *Hallucinated Intermediate Fact* adds a new premise while leaving the goal untouched (“workers take 2 days off, so working days = 23”). *Semantic Drift* adds no premise and changes the goal: every number used is one the problem supplied, but the chain answers a quantity that was not asked for. Diagnostic: strike the hallucinated step and re-run the chain. Under a hallucinated intermediate fact the chain recovers the correct answer; under semantic drift it does not, because the target itself has moved.

C Benchmark Statistics by Hallucination Type

Table 8 reports the number of hallucinated candidates generated per type and task, totalling 12,000 instances across the full benchmark.

D Hallucination Generation Prompts

The following prompts were used to generate hallucinated candidates via GPT-5.4. Each prompt specifies a hallucination method, includes a demonstration, and formats the target instance with field markers. Demonstrations are shown in Bengali for QA and Mathematical Reasoning, and in Latin-script Bangla-English for the Code-Mixed task.

D.1 QA Generation Prompt

Table 9 shows the full QA generation prompt with Bengali demonstrations for all four methods.

D.2 Code-Mixed QA Generation Prompt.

Table 10 mirrors the QA prompt but instructs the model to generate hallucinated answers in Latin-script Bangla-English throughout.

D.3 Summarization Generation Prompt.

Table 11 shows the summarization generation prompt covering three hallucination methods.

D.4 Reasoning Generation Prompt

Table 12 shows the reasoning generation prompt. GPT-5.4 is instructed to modify exactly one step and return output as a structured JSON object.

D.5 Bengali-to-Bangla-English Conversion Prompt

Table 13 shows the Bengali-to-Bangla-English conversion prompt used to produce the code-mixed seed instances from the QA questions.

E Hallucination Evaluation Prompts

The following prompts were used to elicit binary hallucination verdicts (*yes/no*) from the evaluated models. Zero-shot prompts request a single-token response; chain-of-thought prompts require step-by-step reasoning before the final verdict. All prompts are in English except for the input fields, which are in Bengali or Bangla-English depending on the task.

E.1 Zero-Shot Evaluation Prompts

Tables 14, 15, and 16 show the zero-shot evaluation prompts for QA/Code-Mixed QA, Summarization, and Mathematical Reasoning respectively.

E.2 Chain-of-Thought Evaluation Prompts

Under the CoT condition, models are asked to reason through three steps before issuing a verdict. Tables 17, 18, and 19 show the corresponding prompts for QA/Code-Mixed QA, Summarization, and Mathematical Reasoning respectively.

F Full Per-Task Evaluation Results

The following tables extend Table 4 in the main paper by reporting raw verdict counts alongside error rates for all nine models across each of the four tasks. A-err.: Track A error rate on ground-truth instances. B-err.:

Track B error rate on hallucinated candidates. BHS: $\text{BenHalluScore} = \frac{1}{2}(\text{A-err.} + \text{B-err.})$.

F.1 Question Answering

Table 20 shows the full QA evaluation results across all nine models under the dual-track BenHalluScore.

F.2 Summarization

Table 21 shows the full summarization evaluation results across all nine models under the dual-track BenHalluScore.

F.3 Mathematical Reasoning

Table 22 shows the full mathematical reasoning evaluation results across all nine models under the dual-track BenHalluScore.

F.4 Bangla-English Code-Mixed QA

Table 23 shows the full Code-Mixed Banglish QA evaluation results across all nine models under the dual-track BenHalluScore.

F.5 QA Model Error Distribution

Table 20 reports full per-model Track A and Track B error rates on QA. GPT-4.1 mini achieves the best balance (BHS 15.56%), while Mistral-nemo-12B scores worst (BHS 53.84%), driven by its 87.70% Track A error. Models such as LLaMA-3.1-8B show the opposite pattern, with high Track B error and low Track A error, reflecting a systematic “Yes” bias rather than genuine hallucination discrimination.

G BenHalluScore Weight Sensitivity

We recompute $\text{BHS}_w = w_A(\text{A-err.}) + (1 - w_A)(\text{B-err.})$ sweeping $w_A \in \{0.1, \dots, 0.9\}$ for all nine models and all four tasks. Among models that produce content-sensitive verdicts, rankings are highly stable: GPT-4.1 mini tops QA at every weighting from $w_A = 0.2$ to 0.9, and DeepSeek-R1-14B holds the top rank on Mathematical Reasoning from $w_A = 0.2$ through 0.7. At $w_A = 0.1$ the nominal winner is TituLLM-3B on QA, Code-Mixed QA and Mathematical Reasoning - a model that answers “No” on 99-100% of gold instances - and at $w_A = 0.9$ on Mathematical Reasoning, Mistral-nemo-12B (A-err. 1.60%, B-err. 97.00%) scores 11.1, below DeepSeek-R1-14B’s

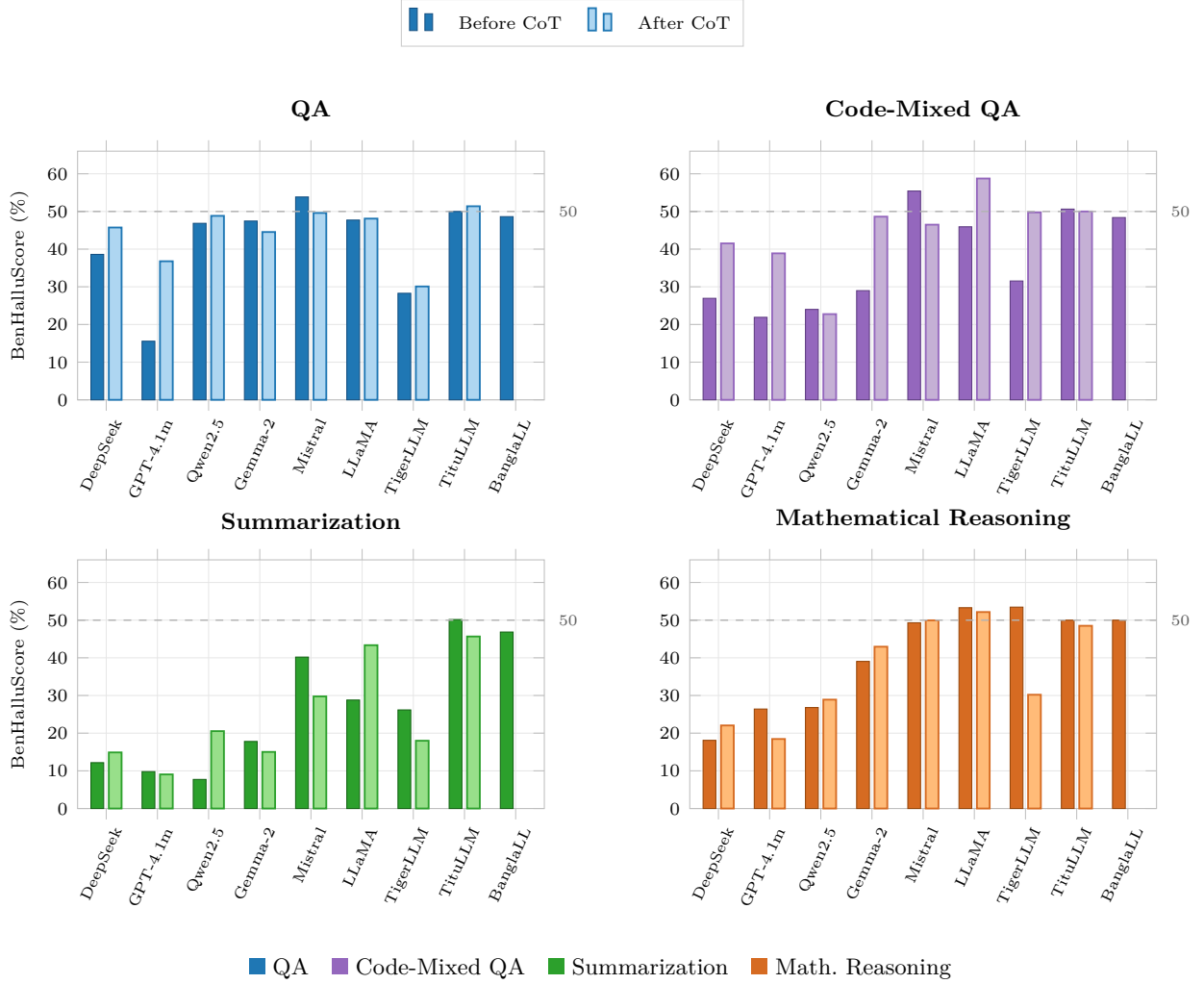


Figure 3: BenHalluScore before and after chain-of-thought prompting across all nine models and all four tasks (32 evaluable combinations; BanglaLLaMA-13B has no *After* bar on any task, having produced only repetition-collapse output under CoT). Each panel shows one task; darker bars = Before CoT, lighter bars = After CoT. Lower is better. The dashed line at 50% marks the score attained by a uniform responder that ignores content; reductions that still land near this line represent a redistribution of response bias rather than a move to a useful operating point (Section 5.4).

19.5, despite flagging almost none of the 1,000 hallucinated chains it was shown.

H Additional Bengali-Centric Models and Expanded CoT

We evaluate TituLLM-3B (Nahin et al., 2025) (hishab/titulm-llama-3.2-3b-v1.1) and BanglaLLaMA-13B (Zehady et al., 2026) (BanglaLLM/bangla-llama-13b-instruct-v0.1) on a stratified 10% sample across all four tasks under the identical dual-track protocol.

I Human Validation Study

Three undergraduate annotators, all native Bengali speakers who have completed writ-

ten and spoken Bangla coursework to the intermediate level, independently annotated a stratified sample spanning all four tasks and all twelve hallucination types, containing both hallucinated candidates and gold instances in benchmark proportions. Annotation was blind to hallucination-type and source labels, and ground truth was set by majority vote (2 of 3).

Code-mixed conversion authenticity. The same annotators reviewed the GPT-generated code-mixed conversions along two axes. Meaning preservation reached Fleiss’ $\kappa = 0.906$, with meaning preserved in 92.9% of scored conversions; the remaining items were

unanimously flagged and have been revised or excluded. Naturalness, rated on a 1–5 Likert scale, averaged 4.38, with 88.9% of conversions rated 4 or higher.

J Seed Selection Analyses

J.1 Cross-Family Judge Check

J.2 BERTScore and Faithfulness Are Independent

Scoring the summarization pool with a multilingual NLI entailment metric (summary against source document) yields a near-zero rank correlation with BERTScore F1 (Spearman $\rho = -0.06$). Selecting the seed pool by NLI instead of BERTScore would change roughly half the documents (495 of 1,000, close to the overlap expected by chance), confirming the two criteria are effectively independent. The same NLI metric separates hallucinated from gold summaries at $AUC = 0.90$, confirming the constructed candidates are genuine faithfulness violations regardless of how the source documents were selected.

J.3 Reasoning Chain Length and Detection Difficulty

Splitting the 1,000 reasoning items at the median chain length and averaging across the evaluated judges, the Track A false-alarm rate rises from 11.3% to 13.1% ($p = 0.014$) and the Track B miss rate from 56.1% to 59.3% ($p = 0.001$), raising BenHalluScore by 2.6 points on the longer half. Comparing the shortest quarter with the longest quarter widens the gap (miss rate 55.5% $\rightarrow$ 60.2%, $p < 0.001$), and BenHalluScore rises monotonically across ten length bins (33.4% $\rightarrow$ 37.6%). Because every item already comes from the longer chains in SOMADHAN (our shortest and longest chains differ in mean length by only $1.72\times$), these figures are a conservative lower bound.

Model	Task	Before CoT			After CoT		
		A-err.	B-err.	BHS	A-err.	B-err.	BHS
DeepSeek-R1-14B	QA	48.00	29.20	38.60	75.05	16.43	45.74 ↑
	Code-Mixed QA	40.10	13.78	26.94	44.00	39.04	41.52 ↑
	Summarization	17.60	6.73	12.17	20.70	9.07	14.89 ↑
	Math. Reasoning	19.90	16.30	18.10	24.40	19.70	22.05 ↑
GPT-4.1 mini	QA	13.90	17.23	15.56	38.90	34.65	36.78 ↑
	Code-Mixed QA	24.00	19.80	21.90	2.00	75.75	38.88 ↑
	Summarization	8.20	11.37	9.79	6.30	11.80	9.05 ↓
	Math. Reasoning	31.70	21.10	26.40	11.60	25.30	18.45 ↓
Qwen2.5-32B	QA	27.90	65.80	46.85	74.40	23.25	48.83 ↑
	Code-Mixed QA	17.40	30.60	24.00	35.00	10.50	22.75 ↓
	Summarization	4.70	10.73	7.72	33.60	7.47	20.54 ↑
	Math. Reasoning	24.80	28.80	26.80	8.70	49.10	28.90 ↑
Gemma-2-27B	QA	16.40	78.50	47.45	50.50	38.58	44.54 ↓
	Code-Mixed QA	19.80	38.15	28.98	51.00	46.25	48.62 ↑
	Summarization	4.20	31.40	17.80	19.00	11.03	15.02 ↓
	Math. Reasoning	5.50	72.60	39.05	3.60	82.30	42.95 ↑
Mistral-nemo-12B	QA	87.70	19.98	53.84	93.30	5.90	49.60 ↓
	Code-Mixed QA	85.90	24.93	55.42	13.13	79.85	46.49 ↓
	Summarization	70.50	9.87	40.19	45.40	14.10	29.75 ↓
	Math. Reasoning	1.60	97.00	49.30	0.10	99.80	49.95 ↑
LLaMA-3.1-8B	QA	13.80	81.63	47.71	83.30	12.90	48.10 ↑
	Code-Mixed QA	13.80	78.05	45.93	19.00	98.50	58.75 ↑
	Summarization	31.80	25.77	28.79	64.90	21.80	43.35 ↑
	Math. Reasoning	17.40	89.20	53.30	28.70	75.60	52.15 ↓
TigerLLM-9B	QA	18.40	38.17	28.28	21.80	38.37	30.09 ↑
	Code-Mixed QA	25.80	37.28	31.54	0.00	99.50	49.75 ↑
	Summarization	4.80	47.53	26.17	17.40	18.57	17.99 ↓
	Math. Reasoning	21.40	85.50	53.45	8.40	52.00	30.20 ↓
TituLLM-3B	QA	99.75	0.25	50.00	63.50	39.25	51.38 ↑
	Code-Mixed QA	99.25	2.00	50.62	34.00	65.99	50.00 ↓
	Summarization	100.00	0.33	50.16	30.00	61.33	45.66 ↓
	Math. Reasoning	100.00	0.00	50.00	66.00	31.00	48.50 ↓
BanglaLLaMA-13B	QA	83.25	14.00	48.62	N/A [‡]		
	Code-Mixed QA	85.50	11.25	48.38	N/A [‡]		
	Summarization	89.00	4.67	46.84	N/A [‡]		
	Math. Reasoning	100.00	0.00	50.00	N/A [‡]		

Table 6: BenHalluScore (BHS) before and after Chain-of-Thought (CoT) prompting across all nine models and all four tasks. A-err. = false alarm rate (%); B-err. = miss rate (%); BHS = $\frac{1}{2}(\text{A-err.} + \text{B-err.})$; lower is better. ■ improved (↓); ■ degraded (↑). [‡]BanglaLLaMA-13B produced no extractable yes/no verdict under CoT on any task, returning repetition-collapse output instead; we report N/A rather than a score, since a numeric value would invite the misreading that BHS = 50 denotes random guessing when the underlying behaviour is systematic degeneration.

Task	Type	Definition	Example
QA	Factualness	Fabricates a concrete fact (name, date, number, place) not present in the context.	Q: When was the Padma Bridge inaugurated? Ground-Truth: 2022 Hallucinated: The Padma Bridge was inaugurated in 2019.
	Comprehension	Misunderstands the question context or intention.	Q: What is the capital of Bangladesh? Ground-Truth: Dhaka Hallucinated: Bangladesh is a South Asian country.
	Specificity	Answers at an inappropriate level of detail.	Q: How many players are in a cricket team? Ground-Truth: 11 Hallucinated: Some.
	Inference	Draws a conclusion not supported by the given information.	Q: Where is Cox’s Bazar located? Ground-Truth: Chittagong Division Hallucinated: Cox’s Bazar is the capital of Bangladesh.
Summ.	Fabricated Content	Introduces a detail not present anywhere in the source.	Context: Patient reports mild fever for two days. Hallucinated: Patient has been hospitalised for a week.
	Non-Factual Addition	Inserts a plausible but unverifiable claim into the summary.	Context: Child has a recurring cough. Hallucinated: Child likely has bronchial asthma.
	Direct Contradiction	Reverses information explicitly stated in the source.	Context: Pain improves when walking. Hallucinated: Pain does not improve with walking.
Mathematical Reasoning	Arithmetic Slip	A single numerical computation is silently altered.	Correct: $25 \times 8 = 200$ hours Hallucinated: $25 \times 8 = 180$ hours
	Formula Misapplication	A related but incorrect operation is substituted.	Correct: $200 \times 5 = 1000$ Hallucinated: $200 + 5 = 205$
	Invalid Deduction	A logically unsupported transition is injected between steps.	Correct: Tax = $20000 \times 0.1 = 2000$ Hallucinated: Tax is split per pair of employees.
	Hallucinated Intermediate Fact	A fabricated assumption is introduced mid-chain as if given.	Correct: Working days = 25 Hallucinated: Workers take 2 days off, so working days = 23.
	Semantic Drift	The reasoning drifts to answer a different question than asked.	Correct: Total cost = wages + tax Hallucinated: Total = wages only = \$20,000.

Table 7: Hallucination type definitions and examples across QA, Summarization, and Mathematical Reasoning. Bengali examples are translated to English for readability.

Task	Hallucination type	Count
QA	Factualness	1,000
	Comprehension	1,000
	Specificity	1,000
	Inference	1,000
Code-mix	Factualness	1,000
	Comprehension	1,000
	Specificity	1,000
	Inference	1,000
Summ.	Fabricated content	1,000
	Non-factual addition	1,000
	Direct contradiction	1,000
Mathematical Reasoning	Arithmetic slip	200
	Formula misapplication	200
	Invalid deduction	200
	Hallucinated intermediate fact	200
	Semantic drift	200
Total		12,000

Table 8: Hallucinated candidate counts by task and type.

[System instruction] I want you act as a hallucination answer generator. The answer should be given in BANGLA . Given a question, right answer, and related knowledge, your objective is to write a hallucinated answer that sounds plausible but is factually incorrect using one of the following methods:
Method 1 — Factualness: You are trying to answer a question but there is a factual contradiction between the answer and the knowledge. You can fabricate some information that does not exist in the provided knowledge.
Demonstration: #Knowledge#: “উইলিয়াম আব্রাহাম সাইমন ওডারল্যান্ড (জন্ম: ৬ ডিসেম্বর, ১৯১৭ — মৃত্যু: ১৮ই মে, ২০০১) ছিলেন একজন ওলন্দাজ-অস্ট্রেলীয় সামরিক কমান্ডো অফিসার...” #Q#: “উইলিয়াম কবে জন্মগ্রহণ করেন?” #Gold#: “৬ ডিসেম্বর, ১৯১৭” #Hallucinated#: “৬ নভেম্বর, ১৯১৬”
Method 2 — Comprehension: You are trying to answer a question but you misunderstand the question context and intention.
Demonstration: #Q#: “স্বাধীন বাংলাদেশের প্রথম চলচ্চিত্রটির নাম কী?” #Gold#: “সুকুমারী” #Hallucinated#: “জহির রায়হান”
Method 3 — Specificity: You are trying to answer a question but the answer is too general or too specific.
Demonstration: #Q#: “কুয়েটের মোট ছাত্রছাত্রীর সংখ্যা কত?” #Gold#: “প্রায় ৬ হাজার” #Hallucinated#: “অজানা”
Method 4 — Inference: You are trying to answer a question but the answer cannot be inferred from the knowledge.
Demonstration: #Q#: “ঢাকার মোট আয়তন কত?” #Gold#: “১৩৪ বর্গমাইল” #Hallucinated#: “২০ মিলিয়ন জনসংখ্যা”
#Hallucinated Answer# can only have about 5 more words than #Right Answer#. #Knowledge#: (context) #Question#: (question) #Right Answer#: (gold) #Hallucinated Answer#: Generate

Table 9: QA hallucination generation prompt. One API call per method per seed instance. The same structure is used for Code-Mixed QA with Bangla-English output and demonstrations.

[System instruction] I want you act as a hallucination answer generator. The answer should be given in CODE-MIXED BANGLA (Bangla-English: Latin-script Bengali) . Given a question, right answer, and related knowledge, write a hallucinated answer that sounds plausible but is factually incorrect using the four methods below.
Method 1 — Factualness: Fabricate information not present in the knowledge.
Demo: #Knowledge#: “William Abraham Simon Ouderland (janma: 6 december, 1917, mrittu: 18 may, 2001) chhilen ekjon commando officer.” #Q#: “Ouderland kobe janmogrohon koren?” #Gold#: “6 december, 1917” #Hallucinated#: “6 november, 1916”
Method 2 — Comprehension: Misunderstand the question context and intention.
Demo: #Q#: “Swadhin Bangladesher prothom cholochchitrotir nam ki?” #Gold#: “Sukumari” #Hallucinated#: “Jahir Raihan”
Method 3 — Specificity: Answer at an inappropriate level of specificity.
Demo: #Q#: “bortomane kuet-er mot chatrochhatrir shongkha koto?” #Gold#: “pray 6 hajar” #Hallucinated#: “ojana”
Method 4 — Inference: Draw a conclusion not supported by the knowledge.
Demo: #Q#: “Dhakar mot ayoton koto?” #Gold#: “134 borgomail” #Hallucinated#: “20 million jonoshon-kha”
#Hallucinated Answer# can only have about 5 more words than #Right Answer#. #Knowledge#: (context) #Question#: (question) #Right Answer#: (gold) #Hallucinated Answer#: Generate

Table 10: Code-Mixed QA hallucination generation prompt. Four methods mirror the QA prompt but use Latin-script Bangla-English throughout.

[System instruction] I want you act as a hallucination summary generator. The summary should be given in BANGLA . Given a document and the right summary, write a hallucinated summary that sounds plausible but is factually incorrect using one of the following methods:
Method 1 — Fabricated Content: You fabricate information that does not exist in the document.
Demo: #Doc#: “আমার মেয়ের বয়স ৫ বছর। সর্দি সমস্যা।” #Gold#: “বয়স ৫, সর্দি সমস্যা, করণীয় কী” #Hallucinated#: “বয়স ৫, জ্বর ও কাশি সহ হাসপাতালে ভর্তি প্রয়োজন”
Method 2 — Non-Factual Addition: Insert information that cannot be inferred from the document.
Demo: #Doc#: “স্ট্রীর বয়স ২২, ডেট ৭ নভেম্বর থেকে ১৪ তারিখে পরিবর্তন।” #Gold#: “স্ট্রীর বয়স ২২, ডেট পরিবর্তন” #Hallucinated#: “ডাক্তার জুলাই মাসে ডেট দিয়ে পরে ১৪ করেছে”
Method 3 — Direct Contradiction: Introduce a clear contradiction between summary and document.
Demo: #Doc#: “হাঁটলে ব্যথা ঠিক হয়ে যায়।” #Gold#: “হাঁটলে ঠিক হয়” #Hallucinated#: “হাঁটলেও ব্যথা কমে না”
#Hallucinated Summary# can only have about 5 more words than #Right Summary#. #Document#: (doc) #Right Summary#: (gold) #Hallucinated Summary#: Generate

Table 11: Summarization hallucination generation prompt. Three methods correspond to fabricated content, non-factual addition, and direct contradiction.

[System instruction]

I want you to act as a hallucinated reasoning chain generator. The output must be in **BANGLA**. Given a math problem, a correct step-by-step reasoning chain, and the correct final answer, your objective is to produce a hallucinated reasoning chain that sounds plausible but is mathematically incorrect. You **MUST** use one of the following hallucination methods:

Method 1 — Arithmetic Slip:

Keep the logic structure completely intact but insert one subtle numerical error in a single calculation step. The wrong number should be close to the correct one (roughly 5% to 20% off). All subsequent steps must use the wrong number consistently. Preserve the `<<expr=result>>` annotation format.

Demonstration:

#Correct step#: “২৫ দিন * ৮ ঘন্টা/দিন = `<<২৫*৮=২০০>>`২০০ ঘন্টা”

#Hallucinated step#: “২৫ দিন * ৮ ঘন্টা/দিন = `<<২৫*৮=১৮০>>`১৮০ ঘন্টা”

Method 2 — Formula Misapplication:

Use a related but wrong operation in one step. The Bengali narrative must justify the wrong operation naturally. All subsequent steps must be consistent with the wrong result.

Demonstration:

#Correct step#: “২০০ ঘন্টা * ৫/ঘন্টা = `<<২০০*৫=১০০০>>`১০০০”

#Hallucinated step#: “২০০ + ৫ = `<<২০০+৫=১০৫>>`১০৫”

Method 3 — Invalid Deduction:

Inject one logically unsupported conclusion in a single step. All subsequent steps must follow from this invalid conclusion.

Demonstration:

#Correct step#: “৳২০,০০০ * .১ = `<<২০০০০*.১=২০০০>>`২,০০০”

#Hallucinated step#: “যেহেতু কর্মচারীর সংখ্যা জোড় (৬ জন), তাই প্রতিটি জুটির জন্য আলাদাভাবে হিসাব করতে হবে। প্রতি জুটির মজুরি = ৳২০,০০০ ÷ ৩ = `<<২০০০০/৩=৬৬৬৬.৬৬৬৬>>`৳৬,৬৬৬।”

Method 4 — Hallucinated Intermediate Fact:

Introduce one fabricated assumption not present in the problem as if it naturally follows. All subsequent steps must proceed from this fabricated fact.

Demonstration:

#Correct step#: “২৫ দিন * ৮ ঘন্টা/দিন = `<<২৫*৮=২০০>>`২০০ ঘন্টা”

#Hallucinated step#: “কমীরা প্রতি মাসে ২ দিন ছুটি ভোগ করেন, তাই কার্যকর কাজের দিন = ২৫ - ২ = `<<২৫-২=২৩>>`২৩ দিন। মোট ঘন্টা = ২৩ * ৮ = `<<২৩*৮=১৮৪>>`১৮৪ ঘন্টা।”

Method 5 — Semantic Drift:

Slightly reinterpret what the problem is asking mid-reasoning so the final answer addresses a different question. The drift must feel like a plausible reading of the problem.

Demonstration:

#Correct step#: “৳২,০০০ + ৳২০,০০০ = `<<২০০০+২০০০০=২২০০০>>`২২,০০০”

#Hallucinated step#: “যেহেতু প্রশ্নটি মূলত মজুরির মোট পরিমাণ জিজ্ঞেস করছে, তাই মোট মজুরি ৳২০,০০০ই চূড়ান্ত উত্তর।”

#Problem#: *<insert Bengali math problem>*

#Correct Reasoning Chain#: *<insert ground-truth CoT>*

#Correct Answer#: *<insert ground-truth answer>*

#Hallucinated Chain#: Generate

Table 12: Hallucination generation prompt for the Mathematical Reasoning task. Five methods correspond to arithmetic slip, formula misapplication, invalid deduction, hallucinated intermediate fact, and semantic drift.

[System instruction] Convert Bengali QA text into natural Bangla-English — Bengali-English code-mix written entirely in Latin script, the way people write on South Asian social media or in casual chats. Output only Latin script - no Bengali characters anywhere. Output only the converted text. No labels, no explanations.

Rules: (1) Romanise Bengali words naturally; use English words where code-mix speakers naturally would. (2) Do NOT translate everything to English; preserve Bengali phrasing where it feels natural. (3) Preserve meaning, tone, names, numbers, dates, brands, and technical terms exactly.

Spelling variation (apply to ~15% of eligible words): common Bangla-English alternates: *bhai/vai*, *bhalo/valo*, *ache/ase*, *ki/kee*, *onek/aneek*, *ektu/aktu*, *shob/sob*, *nai/nei*, *hoise/hoiche*.

Abbreviations (apply sparingly, ~10% of eligible words): *phn* (phone), *msg* (message), *prob* (problem), *pls/plz* (please), *bcz/coz* (because).

Do not: over-abbreviate, use excessive expressive spellings, change factual meaning, or make output hard to read.

[Text to convert]: *<Bengali question and/or passage>*

Table 13: Prompt used to convert the 1,000 QA seed instances from Bengali to Bangla-English. Applied separately to questions and knowledge passages. Ground-Truth answers are retained in their original Bengali form.

You are an evaluator. Decide whether the provided model answer is hallucinated relative to the context and question. Only reply with a single token: **yes** or **no**. No explanation, no punctuation, no extra text. *yes* means the answer contains information not supported by the context or contradicts it. Provide the answer in English only.

Context: $\langle context \rangle$
Question: $\langle question \rangle$
Model answer: $\langle answer \rangle$

Answer now:

Table 14: Zero-shot evaluation prompt for QA and Code-Mixed QA. For code-mixed instances, all fields are in Bangla-English.

You are an evaluator. Decide whether the provided summary is hallucinated relative to the document. Only reply with a single word: **yes** or **no**. *yes* means the summary contains information not supported by or contradicting the document. Provide the answer in English only.

Document: *⟨Bengali medical article⟩*

Summary: *⟨candidate summary⟩*

Answer now (yes or no):

Table 15: Zero-shot evaluation prompt for Summarization.

You are an expert evaluator for Bengali mathematical reasoning tasks. Decide whether the provided reasoning chain is hallucinated. Reply with a single token only: **yes** or **no**. *yes* = chain or answer contains incorrect steps or unsupported facts. English only.

Problem: $\langle \text{Bengali math problem} \rangle$
Reasoning Chain: $\langle \text{CoT solution} \rangle$
Final Answer: $\langle \text{final answer} \rangle$

Answer now:

Table 16: Zero-shot evaluation prompt for Mathematical Reasoning.

You are an evaluator checking whether a model answer is hallucinated.

Question: $\langle question \rangle$ **Model Answer:** $\langle answer \rangle$

Analyze step by step:

Step 1: What factual claims does the answer make?

Step 2: Are these claims supported by or inferable from the question context?

Step 3: Based on steps 1–2, is the answer hallucinated?

Final answer (write only this word on the last line):
yes or no
(yes = hallucinated, no = not hallucinated)

Table 17: Chain-of-thought evaluation prompt for QA and Code-Mixed QA.

You are an evaluator checking whether a summary is hallucinated relative to a document.

Document: $\langle document \rangle$ **Summary:** $\langle summary \rangle$

Analyze step by step:

Step 1: List the key claims made in the summary.

Step 2: For each claim, check whether it is directly supported by the document.

Step 3: Based on steps 1–2, decide your final answer.

Final answer (write only this word on the last line):
yes or no
(yes = hallucinated, no = not hallucinated)

Table 18: Chain-of-thought evaluation prompt for Summarization.

You are an expert evaluator for Bengali mathematical reasoning tasks.

Question: $\langle problem \rangle$ Reasoning Chain: $\langle CoT \rangle$

Answer: $\langle final\ answer \rangle$

Analyze step by step:

Step 1: Is each calculation or logical step mathematically correct?

Step 2: Does the final answer follow logically from the reasoning chain?

Step 3: Based on steps 1–2, is this reasoning chain hallucinated?

Respond **ONLY** with a JSON object on the last line:

```
{"is_hallucinated": "Yes"}
```

 or

```
{"is_hallucinated": "No"}
```

Table 19: Chain-of-thought evaluation prompt for Mathematical Reasoning.

#	Model	Track A — Ground Truth (1,000)			Track B — Hallucinated (4,000)			BHS
		Correct	Wrong	A-err.	Correct	Wrong	B-err.	
1	Qwen2.5-32B-Instruct	771	279	27.90%	1,365	2,632	65.80%	46.85%
2	Gemma-2-27B	836	164	16.40%	860	3,140	78.50%	47.45%
3	DeepSeek-R1-14B	342	480	48.00%	2,820	1,168	29.20%	38.60%
4	GPT-4.1 mini	861	139	13.90%	3,311	689	17.23%	15.56%
5	Mistral-nemo-12B	123	877	87.70%	3,201	799	19.98%	53.84%
6	LLaMA-3.1-8B-Instruct	862	138	13.80%	735	3,265	81.63%	47.71%
7	TigerLLM-9B	816	184	18.40%	2,855	1,145	38.17%	28.28%

Table 20: QA full results. **Bold** = best and worst BHS.

#	Model	Track A — Ground Truth (1,000)			Track B — Hallucinated (3,000)			BHS
		Correct	Wrong	A-err.	Correct	Wrong	B-err.	
1	Qwen2.5-32B-Instruct	953	47	4.70%	2,678	322	10.73%	7.72%
2	Gemma-2-27B	958	42	4.20%	2,058	942	31.40%	17.80%
3	DeepSeek-R1-14B	824	176	17.60%	2,798	202	6.73%	12.17%
4	GPT-4.1 mini	918	82	8.20%	2,659	341	11.37%	9.79%
5	Mistral-nemo-12B	295	705	70.50%	2,704	296	9.87%	40.19%
6	LLaMA-3.1-8B-Instruct	682	318	31.80%	2,227	773	25.77%	28.79%
7	TigerLLM-9B	952	48	4.80%	1,574	1,426	47.53%	26.17%

Table 21: Summarization full results. **Bold** = best and worst BHS.

#	Model	Track A — Ground Truth (1,000)			Track B — Hallucinated (1,000)			BHS
		Correct	Wrong	A-err.	Correct	Wrong	B-err.	
1	Qwen2.5-32B-Instruct	752	248	24.80%	703	288	28.80%	26.80%
2	Gemma-2-27B	945	55	5.50%	274	726	72.60%	39.05%
3	DeepSeek-R1-14B	801	199	19.90%	837	163	16.30%	18.10%
4	GPT-4.1 mini	683	317	31.70%	789	211	21.10%	26.40%
5	Mistral-nemo-12B	984	16	1.60%	30	970	97.00%	49.30%
6	LLaMA-3.1-8B-Instruct	826	174	17.40%	108	892	89.20%	53.30%
7	TigerLLM-9B	786	214	21.40%	145	855	85.50%	53.45%

Table 22: Mathematical Reasoning full results. **Bold** = best and worst BHS.

#	Model	Track A — Ground Truth (1,000)			Track B — Hallucinated (4,000)			BHS
		Correct	Wrong	A-err.	Correct	Wrong	B-err.	
1	Qwen2.5-32B-Instruct	826	174	17.40%	2,776	1,224	30.60%	24.00%
2	Gemma-2-27B	802	198	19.80%	2,474	1,526	38.15%	28.98%
3	DeepSeek-R1-14B	596	401	40.10%	3,439	551	13.78%	26.94%
4	GPT-4.1 mini	760	240	24.00%	3,208	792	19.80%	21.90%
5	Mistral-nemo-12B	141	859	85.90%	3,003	997	24.93%	55.42%
6	LLaMA-3.1-8B-Instruct	862	138	13.80%	878	3,122	78.05%	45.93%
7	TigerLLM-9B	739	258	25.80%	2,496	1,491	37.28%	31.54%

Table 23: Bangla-English Code-Mixed QA full results. **Bold** = best and worst BHS.

Task	Model	w_A								
		0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
QA	DeepSeek-R1-14B	31.1	33.0	34.8	36.7	38.6	40.5	42.4	44.2	46.1
	GPT-4.1 mini	16.9	16.6	16.2	15.9	15.6	15.2	14.9	14.6	14.2
	Qwen2.5-32B	62.0	58.2	54.4	50.6	46.8	43.1	39.3	35.5	31.7
	Gemma-2-27B	72.3	66.1	59.9	53.7	47.5	41.2	35.0	28.8	22.6
	Mistral-nemo-12B	26.8	33.5	40.3	47.1	53.8	60.6	67.4	74.2	80.9
	LLaMA-3.1-8B	74.8	68.1	61.3	54.5	47.7	40.9	34.1	27.4	20.6
	TigerLLM-9B	36.2	34.2	32.2	30.3	28.3	26.3	24.3	22.4	20.4
	TituLLM-3B	10.2	20.2	30.1	40.1	50.0	59.9	69.9	79.9	89.8
	BanglaLLaMA-13B	20.9	27.9	34.8	41.7	48.6	55.5	62.5	69.4	76.3
Code-Mix	DeepSeek-R1-14B	16.4	19.0	21.7	24.3	26.9	29.6	32.2	34.8	37.5
	GPT-4.1 mini	20.2	20.6	21.1	21.5	21.9	22.3	22.7	23.2	23.6
	Qwen2.5-32B	29.3	28.0	26.6	25.3	24.0	22.7	21.4	20.0	18.7
	Gemma-2-27B	36.3	34.5	32.6	30.8	29.0	27.1	25.3	23.5	21.6
	Mistral-nemo-12B	31.0	37.1	43.2	49.3	55.4	61.5	67.6	73.7	79.8
	LLaMA-3.1-8B	71.6	65.2	58.8	52.4	45.9	39.5	33.1	26.6	20.2
	TigerLLM-9B	36.1	35.0	33.8	32.7	31.5	30.4	29.2	28.1	26.9
	TituLLM-3B	11.7	21.5	31.2	40.9	50.6	60.3	70.1	79.8	89.5
	BanglaLLaMA-13B	18.7	26.1	33.5	41.0	48.4	55.8	63.2	70.7	78.1
Summ.	DeepSeek-R1-14B	7.8	8.9	10.0	11.1	12.2	13.3	14.3	15.4	16.5
	GPT-4.1 mini	11.1	10.7	10.4	10.1	9.8	9.5	9.2	8.8	8.5
	Qwen2.5-32B	10.1	9.5	8.9	8.3	7.7	7.1	6.5	5.9	5.3
	Gemma-2-27B	28.7	26.0	23.2	20.5	17.8	15.1	12.4	9.6	6.9
	Mistral-nemo-12B	15.9	22.0	28.1	34.1	40.2	46.2	52.3	58.4	64.4
	LLaMA-3.1-8B	26.4	27.0	27.6	28.2	28.8	29.4	30.0	30.6	31.2
	TigerLLM-9B	43.3	39.0	34.7	30.4	26.2	21.9	17.6	13.3	9.1
	TituLLM-3B	10.3	20.3	30.2	40.2	50.2	60.1	70.1	80.1	90.0
	BanglaLLaMA-13B	13.1	21.5	30.0	38.4	46.8	55.3	63.7	72.1	80.6
Math. Reas.	DeepSeek-R1-14B	16.7	17.0	17.4	17.7	18.1	18.5	18.8	19.2	19.5
	GPT-4.1 mini	22.2	23.2	24.3	25.3	26.4	27.5	28.5	29.6	30.6
	Qwen2.5-32B	28.4	28.0	27.6	27.2	26.8	26.4	26.0	25.6	25.2
	Gemma-2-27B	65.9	59.2	52.5	45.8	39.0	32.3	25.6	18.9	12.2
	Mistral-nemo-12B	87.5	77.9	68.4	58.8	49.3	39.8	30.2	20.7	11.1
	LLaMA-3.1-8B	82.0	74.8	67.7	60.5	53.3	46.1	38.9	31.8	24.6
	TigerLLM-9B	79.1	72.7	66.3	59.9	53.5	47.0	40.6	34.2	27.8
	TituLLM-3B	10.0	20.0	30.0	40.0	50.0	60.0	70.0	80.0	90.0
	BanglaLLaMA-13B	10.0	20.0	30.0	40.0	50.0	60.0	70.0	80.0	90.0

Table 24: Nine-point weight sweep of $BHS_w = w_A(\text{A-err.}) + (1 - w_A)(\text{B-err.})$ for every model and task (zero-shot). **Bold** = best (lowest) model at that weighting; TituLLM-3B and BanglaLLaMA-13B tie for best on Mathematical Reasoning at $w_A = 0.1$. Rankings are stable through the central range; at the extremes the nominal winner is in almost every case a degenerate responder.

Task	Items	Fleiss' κ	Exact	Unanim.
QA	241/250	0.918	97.2%	95.9%
Code-Mixed QA	250/250	0.914	98.4%	97.6%
Summarization	200/200	0.926	97.3%	96.0%
Math. Reasoning	300/300	0.911	97.3%	96.0%

Table 25: Inter-annotator agreement. Nine QA items were excluded because one annotator skipped them. By the Landis–Koch interpretation, $\kappa = 0.911$ – 0.926 is “almost perfect” agreement.

Generator	Agreement	Differs
Qwen2.5-14B	476/500 (95.2%)	24
DeepSeek-R1-14B	462/500 (92.4%)	38
Gemma-2-9B	456/500 (91.2%)	44

Table 26: Agreement between the Qwen2.5-32B judge and an independent GPT-4.1 mini judge on a random 500-instance subsample (seed 42), identical prompt. Disagreement is lowest for the same-family generator, the opposite of a self-preference effect.